

HTAN: Hyper TransAttUNet with Manifold-Constrained Hyper-Connections for Medical Image Segmentation

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Abstract—Accurate medical image segmentation is essential for reliable disease diagnosis, lesion analysis, and organ morphometry. U-Net-based encoder-decoder architectures have achieved strong performance in this task by combining contextual encoding with spatial reconstruction. Recent attention-guided variants further improve feature representation by modeling long-range dependencies and aggregating multi-scale decoder information. However, the bottleneck representation in these models is still limited by single-stream feature processing, which restricts the interaction of high-level semantic features before decoding. In this paper, we propose HTAN, Hyper TransAttUNet, a medical image segmentation architecture that integrates Manifold-Constrained Hyper-Connections into the TransAttUNet bottleneck. The proposed design expands the Self-Aware Attention bottleneck into multiple residual streams and constrains their mixing using doubly stochastic matrices obtained through Sinkhorn-Knopp projection. This enables richer bottleneck feature interaction while preserving stable information flow. We evaluate HTAN on three medical segmentation benchmarks: ISIC-2018 skin lesion segmentation, GlaS gland segmentation, and Data Science Bowl nuclei segmentation. Experimental results show consistent improvements over the reproduced TransAttUNet_R baseline, with Dice gains of 1.05%, 3.57%, and 1.07% on ISIC-2018, GlaS, and Bowl, respectively. These results demonstrate that manifold-constrained multi-stream bottleneck processing is an effective way to strengthen TransAttUNet for medical image segmentation.

Index Terms—Medical image segmentation, TransAttUNet, U-Net, self-aware attention, hyper-connections, manifold-constrained hyper-connections, Sinkhorn-Knopp.

I. INTRODUCTION

Medical image segmentation is a fundamental task in medical image analysis, where anatomical structures or pathological regions are delineated at the pixel level. Accurate segmentation is required in many clinical workflows because downstream decisions often depend on the shape, size, and location of the segmented region. In radiotherapy, for example, segmentation is used to contour target volumes and organs-at-risk before treatment planning. Recent clinical studies show that automatic segmentation is increasingly being evaluated and adopted in real medical centers. Deep learning-based auto-segmentation has been implemented at two cancer centers for radiotherapy planning, where contours were generated for approximately

551 eligible cases [1]. A prospective multicenter trial involving 500 patients and five centers further evaluated AI-assisted organ-at-risk delineation in thoracic and breast cancer radiotherapy [2]. Survey evidence also suggests growing clinical adoption, with 65.7% of respondents in German-speaking radiotherapy centers reporting routine use of AI-based auto-contouring software [3]. These findings highlight the practical importance of developing accurate and reliable segmentation models.

Deep learning has played a central role in advancing automatic medical image segmentation. Among existing architectures, U-Net has become a foundational model because it follows an encoder-decoder structure designed for pixel-level prediction [5]. The encoder extracts increasingly abstract image features, while the decoder reconstructs the final segmentation mask. Skip connections transfer high-resolution encoder features to the decoder, helping recover spatial details that may be weakened during downsampling. This design makes U-Net effective for biomedical segmentation and has motivated many later encoder-decoder segmentation networks.

Despite this progress, U-Net-style convolutional architectures still have important limitations. Convolutional operations mainly aggregate information from local neighborhoods, which can restrict the ability of the model to capture long-range contextual dependencies. This is problematic in medical segmentation because accurate boundary delineation may depend not only on local texture, but also on broader spatial and semantic relationships across the image. In addition, conventional skip connections usually operate between corresponding encoder and decoder stages at the same scale, without fully modeling relationships among features from different decoder levels [4].

TransAttUNet addresses these limitations by enhancing the U-shaped architecture with attention and multi-scale decoding [4]. It introduces a Self-Aware Attention (SAA) module at the bridge between the encoder and decoder. This module combines Transformer Self-Attention (TSA), which captures long-range feature interactions, and Global Spatial Attention (GSA), which aggregates global spatial context. TransAttUNet also uses multi-scale skip connections in the decoder to com-

bine feature representations from different semantic scales. Together, these components improve the ability of the network to model both global context and local spatial details.

However, although TransAttUNet enriches the bottleneck feature using the Self-Aware Attention module, the resulting representation is still propagated through a single residual stream before decoding [4]. This creates a representational bottleneck, where different high-level semantic cues extracted by the attention module must be carried and updated through the same feature path, limiting explicit interaction among complementary feature streams. A natural way to increase the expressive capacity of this bottleneck is to expand the residual representation into multiple streams, allowing different feature components to be mixed and reused before reconstruction. However, unconstrained multi-stream residual mixing can disrupt the identity-mapping property of residual connections and may lead to unstable signal propagation. Manifold-Constrained Hyper-Connections (mHC) address this issue by expanding the residual stream while constraining the residual mixing matrix to be doubly stochastic through Sinkhorn-Knopp projection [6]. This motivates our use of mHC in the TransAttUNet bottleneck, where richer multi-stream feature interaction can be introduced while maintaining stable information flow.

In this paper, we propose HTAN, Hyper TransAttUNet, a medical image segmentation architecture that integrates Manifold-Constrained Hyper-Connections into the TransAttUNet bottleneck. Instead of relying only on a single-stream bottleneck representation, HTAN expands the Self-Aware Attention bottleneck into multiple residual streams and constrains their mixing through the mHC formulation. This design enables richer bottleneck feature interaction while preserving stable residual propagation.

The main contributions of this work are summarized as follows:

- We propose HTAN, a hyper-connected extension of TransAttUNet that integrates Manifold-Constrained Hyper-Connections into the Self-Aware Attention bottleneck for medical image segmentation.
- We introduce a stable multi-stream bottleneck representation, where residual feature streams are mixed through doubly stochastic matrices obtained by Sinkhorn-Knopp projection. This allows richer feature interaction while preserving stable information flow.
- We evaluate HTAN on three medical image segmentation benchmarks: ISIC-2018 skin lesion segmentation, GlaS gland segmentation, and Data Science Bowl nuclei segmentation. Across these datasets, HTAN consistently improves over the reproduced TransAttUNet_R baseline, achieving Dice gains of 1.05%, 3.57%, and 1.07%, respectively.
- We provide ablation studies on the Hres-only variant, the residual stream expansion rate, and the number of mHC blocks, showing how each design choice affects segmentation performance and model stability.

II. RELATED WORK

A. Encoder-Decoder Networks for Medical Image Segmentation

Encoder-decoder networks have been widely used for medical image segmentation because they can transform an input image into a dense pixel-level prediction. U-Net is one of the most influential architectures in this direction. It consists of a contracting path that captures contextual information and an expanding path that reconstructs spatial resolution for precise localization [5]. The skip connections between corresponding encoder and decoder stages allow high-resolution features from the encoder to be reused during decoding, helping preserve spatial details that may be weakened by downsampling. This design has made U-Net a common foundation for later biomedical segmentation networks.

Several later models extended the U-shaped design by improving feature reuse, contextual aggregation, or decoder reconstruction. For example, multi-scale context models such as UNet++ and related U-Net variants redesign skip pathways to better combine features across different semantic levels [15]. These works show that segmentation performance depends not only on the encoder representation, but also on how features are transferred and reconstructed in the decoder. However, convolution-based encoder-decoder networks are still limited in modeling long-range dependencies, since their feature extraction is mainly driven by local convolutional operations.

B. Attention and Transformer-Based Medical Segmentation

Attention mechanisms have been introduced into U-Net-style architectures to help the network focus on more relevant regions and feature responses. Attention U-Net uses attention gates to suppress irrelevant feature activations and highlight salient structures during decoding [9]. Channel-UNet further explores spatial and channel-wise feature recalibration to improve segmentation representation [16]. These attention-guided models demonstrate that selectively weighting feature responses can improve the ability of encoder-decoder networks to segment clinically relevant regions.

Transformer-based methods further address the limitation of local convolution by modeling broader contextual relationships. TransUNet combines Transformer-based encoding with U-Net-style decoding for medical image segmentation [14]. Other transformer-based segmentation methods, including Swin-Unet, SegFormer, and MCTrans, also explore different ways of using self-attention or transformer representations for dense prediction tasks [11]–[13]. Compared with purely convolutional models, these approaches improve the ability to capture long-range dependencies and global semantic relationships.

TransAttUNet builds on this direction by introducing a Self-Aware Attention (SAA) module at the bridge between the encoder and decoder [4]. The SAA module combines Transformer Self-Attention (TSA), which models long-range feature interactions, and Global Spatial Attention (GSA), which captures global spatial context. This makes TransAttUNet a

strong attention-guided U-shaped baseline for medical image segmentation. In this work, we use TransAttUNet_R as the direct baseline because it combines the SAA bottleneck with residual multi-scale decoder connections.

C. Multi-Scale Feature Fusion in U-Shaped Networks

Medical image segmentation often requires both high-level semantic information and fine spatial detail. Multi-scale feature fusion has therefore become an important design principle in U-shaped networks. Standard U-Net transfers features from the encoder to the decoder at corresponding resolutions, but this same-scale transfer does not fully exploit relationships among decoder features at different semantic levels. Methods such as UNet++ redesign skip connections to aggregate multi-scale features more effectively [15].

TransAttUNet further studies multi-scale skip connections inside the decoder [4]. It compares cascade, residual, and dense decoder connections. In the residual decoder connection, features are progressively combined between decoder stages, allowing information to grow step by step during reconstruction. In the dense decoder connection, features from earlier decoder blocks are reused by later blocks. These designs help reduce information loss during upsampling and allow the decoder to combine features from different semantic scales. Among the TransAttUNet variants, the residual version, TransAttUNet_R, is especially relevant to this work because it provides the baseline architecture into which we integrate manifold-constrained hyper-connections.

D. Residual Stream Expansion and Manifold-Constrained Hyper-Connections

Residual connections are important for stable information propagation in deep networks because they preserve an identity mapping path across layers. Hyper-Connections extend this idea by expanding a single residual stream into multiple residual streams and introducing learnable mappings to read from, write to, and mix these streams [7]. This expanded residual representation increases the flexibility of feature interaction by allowing information to move across multiple streams instead of being carried through a single path.

However, unconstrained residual stream mixing can be unstable. The mHC paper shows that the residual mixing matrix in Hyper-Connections may disrupt the identity-mapping property and cause signal amplification or attenuation across layers [6]. Manifold-Constrained Hyper-Connections address this issue by constraining the residual mixing matrix to be doubly stochastic through Sinkhorn-Knopp projection. This constraint keeps stream mixing stable while still allowing information exchange among residual streams.

Existing mHC work studies this idea in the context of large-scale model training. In contrast, this paper investigates whether the same principle of stable residual stream expansion can improve a medical image segmentation architecture. Specifically, we integrate mHC into the Self-Aware Attention bottleneck of TransAttUNet, allowing multi-stream bottleneck

feature interaction while preserving stable residual propagation.

III. METHODOLOGY

A. Overview of the Proposed HTAN Architecture

The proposed HTAN architecture builds on TransAttUNet_R, a U-shaped encoder-decoder segmentation model that combines a Self-Aware Attention (SAA) bottleneck with residual multi-scale decoder connections [4]. Given an input medical image $X \in \mathbb{R}^{B \times C \times H \times W}$, the network predicts a pixel-level segmentation mask $\hat{Y} \in \mathbb{R}^{B \times 1 \times H \times W}$. The encoder extracts hierarchical image features, the bottleneck refines the deepest semantic representation, and the decoder reconstructs the final segmentation mask.

HTAN modifies the TransAttUNet bottleneck by integrating Manifold-Constrained Hyper-Connections (mHC) into the SAA feature-processing stage. Instead of passing the SAA bottleneck representation through a single residual stream, HTAN expands it into n residual streams, mixes these streams using a constrained residual mapping, and then fuses the resulting representation before decoding. The constrained stream mixing follows the mHC formulation, where the residual mapping is projected onto the Birkhoff polytope through Sinkhorn-Knopp normalization [6]. Fig. 1 illustrates the overall HTAN architecture.

B. TransAttUNet Baseline

1) *Encoder-Decoder Backbone*: TransAttUNet follows a standard U-shaped encoder-decoder architecture for medical image segmentation [4]. The encoder progressively reduces spatial resolution while increasing semantic abstraction, and the decoder gradually restores spatial resolution to generate the segmentation mask. This design is inherited from U-Net, where the contracting path captures context and the expanding path enables precise localization [5].

Let F_i denote the encoder feature at stage i . The encoder applies repeated convolution and downsampling operations to produce hierarchical feature maps:

$$F_i = E_i(F_{i-1}), \quad i = 1, 2, \dots, L, \quad (1)$$

where $E_i(\cdot)$ represents the convolutional operations and downsampling at encoder stage i . The deepest encoder feature F_L is passed to the bottleneck, where TransAttUNet applies the SAA module.

2) *Self-Aware Attention Module*: The SAA module is positioned at the bridge between the encoder and decoder in TransAttUNet [4]. It contains two complementary attention branches: Transformer Self-Attention (TSA) and Global Spatial Attention (GSA). TSA models long-range feature interactions among encoder features, while GSA captures global spatial context. The SAA output is obtained by combining the TSA feature, the GSA feature, and the original encoder feature:

$$F_{\text{SAA}} = \lambda_1 F_{\text{TSA}} + \lambda_2 F_{\text{GSA}} + F_{\text{en}}, \quad (2)$$

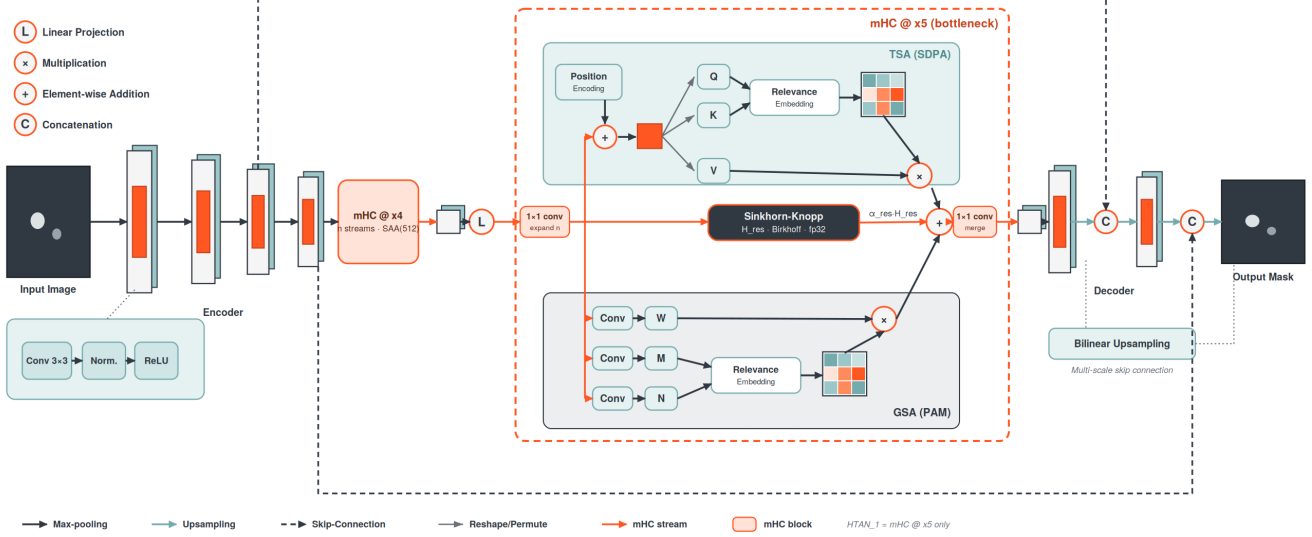


Fig. 1. Overview of the proposed HTAN architecture. The encoder-decoder backbone of TransAttUNet_R is augmented with mHC blocks at the two deepest encoder levels (x_4 and x_5). At the bottleneck (x_5), the mHC module expands the SAA representation into n parallel residual streams mixed via Sinkhorn-Knopp projection before decoding. HTAN_1 applies mHC only at x_5 ; HTAN_2 applies mHC at both x_4 and x_5 .

where F_{en} is the input encoder feature, F_{TSA} and F_{GSA} are the outputs of the TSA and GSA branches, and λ_1 and λ_2 control the contribution of the two attention branches.

3) *Transformer Self-Attention*: The TSA branch follows the multi-head self-attention mechanism used in Transformer-based models [4]. Given an encoder feature $F \in \mathbb{R}^{c \times h \times w}$, the feature is projected into query, key, and value representations:

$$Q = FW_q, \quad K = FW_k, \quad V = FW_v, \quad (3)$$

where W_q , W_k , and W_v are learnable projection matrices. The attention response is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (4)$$

where d_k is the key dimension. This operation allows the bottleneck feature to capture non-local dependencies among spatial positions.

4) *Global Spatial Attention*: The GSA branch is designed to capture global spatial relationships in the encoder feature map [4]. While TSA models pairwise self-attention among encoded features, GSA aggregates global spatial context and produces a complementary attention-enhanced representation. The two branches are therefore used together in the SAA module to strengthen the bottleneck representation before decoding.

5) *Multi-Scale Residual Skip Connections*: TransAttUNet also introduces multi-scale skip connections between decoder blocks to aggregate features from different semantic scales [4]. In the residual decoder variant, TransAttUNet_R, decoder features are progressively combined through residual step-growth connections. This design reduces information loss during upsampling and allows the decoder to reuse contextual

information from deeper decoder stages. Since HTAN is built on TransAttUNet_R, the same residual multi-scale decoder structure is preserved in our model.

C. Manifold-Constrained Hyper-Connections

1) *Residual Stream Expansion*: A standard residual connection propagates information through a single stream:

$$x_{l+1} = x_l + F_l(x_l), \quad (5)$$

where x_l is the input to layer l and $F_l(\cdot)$ is the layer transformation. Hyper-Connections extend this idea by expanding the residual representation into n streams and introducing learnable mappings to read from, write to, and mix these streams [6]. This allows the model to support richer interaction among residual feature streams.

In the mHC formulation, the multi-stream residual state is denoted as:

$$X_l \in \mathbb{R}^{B \times n \times D}, \quad (6)$$

where B is the batch size, n is the number of residual streams, and D is the flattened feature dimension. The goal is to update the residual streams while preserving stable information propagation.

2) *Pre-Mapping, Post-Mapping, and Residual Mapping*: mHC uses three learnable mappings: H_{pre} , H_{post} , and H_{res} [6]. The pre-mapping H_{pre} combines information from the residual streams before applying the wrapped transformation. The post-mapping H_{post} writes the transformed feature back into the residual streams. The residual mapping H_{res} mixes the residual streams directly:

$$X_{l+1} = H_{\text{res}}^{(l)} X_l + \left(H_{\text{post}}^{(l)}\right)^T F_l\left(H_{\text{pre}}^{(l)} X_l\right). \quad (7)$$

In HTAN, the wrapped transformation $F_l(\cdot)$ corresponds to the SAA-based bottleneck processing adapted to the mHC stream representation.

3) *Doubly Stochastic Constraint*: Unconstrained residual stream mixing can disrupt the identity-mapping property of residual connections and may cause unstable signal propagation [6]. To address this, mHC constrains the residual mapping H_{res} to be doubly stochastic. A doubly stochastic matrix has non-negative entries, rows that sum to one, and columns that sum to one:

$$H_{\text{res}} \mathbf{1}_n = \mathbf{1}_n, \quad \mathbf{1}_n^T H_{\text{res}} = \mathbf{1}_n^T, \quad H_{\text{res}} \geq 0, \quad (8)$$

where $\mathbf{1}_n$ is an n -dimensional vector of ones. The set of such matrices forms the Birkhoff polytope. This constraint allows stream interaction while preserving stable forward and backward propagation [6].

4) *Sinkhorn-Knopp Projection*: To enforce the doubly stochastic constraint, mHC applies Sinkhorn-Knopp normalization to the residual mapping [6]. Given an unconstrained matrix Z , a positive matrix is first obtained:

$$S^{(0)} = \exp(Z). \quad (9)$$

Then, row and column normalization are applied iteratively:

$$S_{ij}^{(t+\frac{1}{2})} = S_{ij}^{(t)} / \sum_{j'} S_{ij'}^{(t)}, \quad (10)$$

$$S_{ij}^{(t+1)} = S_{ij}^{(t+\frac{1}{2})} / \sum_{i'} S_{i'j}^{(t+\frac{1}{2})}. \quad (11)$$

After a fixed number of iterations, the resulting matrix approximates a doubly stochastic residual mapping. In our implementation, the Sinkhorn-Knopp operation is computed in FP32 to improve numerical stability during mixed-precision training.

D. Integration of mHC into TransAttUNet

1) *Reshaping 2D Image Features into mHC Streams*: The SAA module in TransAttUNet operates on two-dimensional encoder feature maps, while mHC operates on residual streams [4], [6]. Therefore, HTAN introduces a reshaping step to bridge these representations. Given the bottleneck encoder feature:

$$F_{\text{en}} \in \mathbb{R}^{B \times C \times H' \times W'}, \quad (12)$$

we flatten the spatial and channel dimensions into:

$$z = \text{vec}(F_{\text{en}}) \in \mathbb{R}^{B \times D}, \quad D = CH'W'. \quad (13)$$

The flattened feature is then expanded into n residual streams:

$$X^{(0)} = \text{StreamInit}(z) \in \mathbb{R}^{B \times n \times D}. \quad (14)$$

2) *mHC-Enhanced SAA Bottleneck*: After stream initialization, the mHC module updates the bottleneck representation through constrained residual stream mixing. For an HTAN variant with K mHC blocks, the update is:

$$X^{(k+1)} = \text{mHC}_k(X^{(k)}, F_{\text{SAA}}), \quad k = 0, 1, \dots, K-1. \quad (15)$$

Here, F_{SAA} denotes the SAA transformation adapted to the flattened residual-stream representation. This design allows the attention-enhanced bottleneck feature to be processed through multiple stable residual streams instead of a single feature path.

3) *Stream Fusion and Decoder Reconstruction*: After the mHC-enhanced bottleneck processing, the residual streams are fused back into a single bottleneck vector:

$$z_{\text{HTAN}} = \text{StreamFuse}(X^{(K)}). \quad (16)$$

The fused representation is reshaped back into a two-dimensional feature map:

$$F_{\text{HTAN}} = \text{reshape}(z_{\text{HTAN}}) \in \mathbb{R}^{B \times C \times H' \times W'}. \quad (17)$$

The resulting feature map is then passed to the TransAttUNet_R decoder, which reconstructs the segmentation mask using residual multi-scale decoder connections [4].

Fig. 2 illustrates the internal structure of a single mHC block as used in HTAN_1, showing the stream expansion, SAA processing, Sinkhorn-Knopp residual path, and stream merge. Fig. 3 shows the two-block HTAN_2 configuration where mHC is applied sequentially at x_4 and x_5 .

E. HTAN Architectural Variants

We evaluate several HTAN variants to study the effect of residual stream expansion and the number of mHC blocks. Let n denote the number of residual streams and K denote the number of mHC blocks. The main variants are summarized in Table I.

TABLE I
HTAN ARCHITECTURAL VARIANTS.

Variant	K	n	Description
HTAN_1 Hres-only	1	2	Constrains H_{res} only
HTAN_1 $n = 2$	1	2	One mHC block with two streams
HTAN_1 $n = 4$	1	4	One mHC block with four streams
HTAN_2 $n = 2$	2	2	Two mHC blocks with two streams
HTAN_2 $n = 4$	2	4	Two mHC blocks with four streams

The Hres-only model is used as an ablation to isolate the effect of constraining only the residual stream mixing matrix. The $n = 2$ and $n = 4$ variants test whether increasing the number of residual streams improves bottleneck feature interaction. The HTAN_1 and HTAN_2 variants evaluate whether using one or two mHC blocks provides a better balance between performance and model complexity.

HTAN_1 — mHC block

Manifold-Constrained Hyper-Connection · single block at x₅ (bottleneck) · n = 2

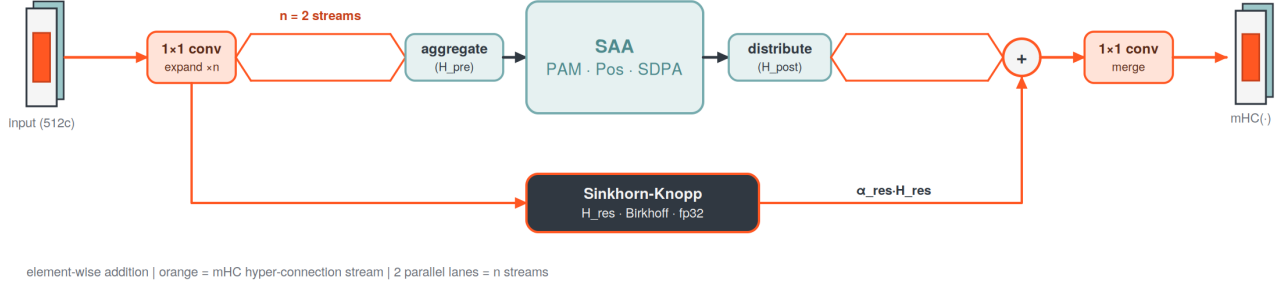


Fig. 2. Internal structure of the mHC block used in HTAN_1. The input feature is expanded into $n = 2$ residual streams via a 1×1 convolution. The streams are aggregated by H_{pre} , passed through the SAA module (PAM, positional encoding, SDPA), and distributed back by H_{post} . In parallel, the residual stream mixing matrix H_{res} is projected onto the Birkhoff polytope via Sinkhorn-Knopp (fp32). The two paths are combined by element-wise addition, and the streams are merged back by a 1×1 convolution.

HTAN_2 — two mHC blocks (mHC @ x₄ → mHC @ x₅)

Manifold-Constrained Hyper-Connections at the two deepest encoder levels · n = 2 streams

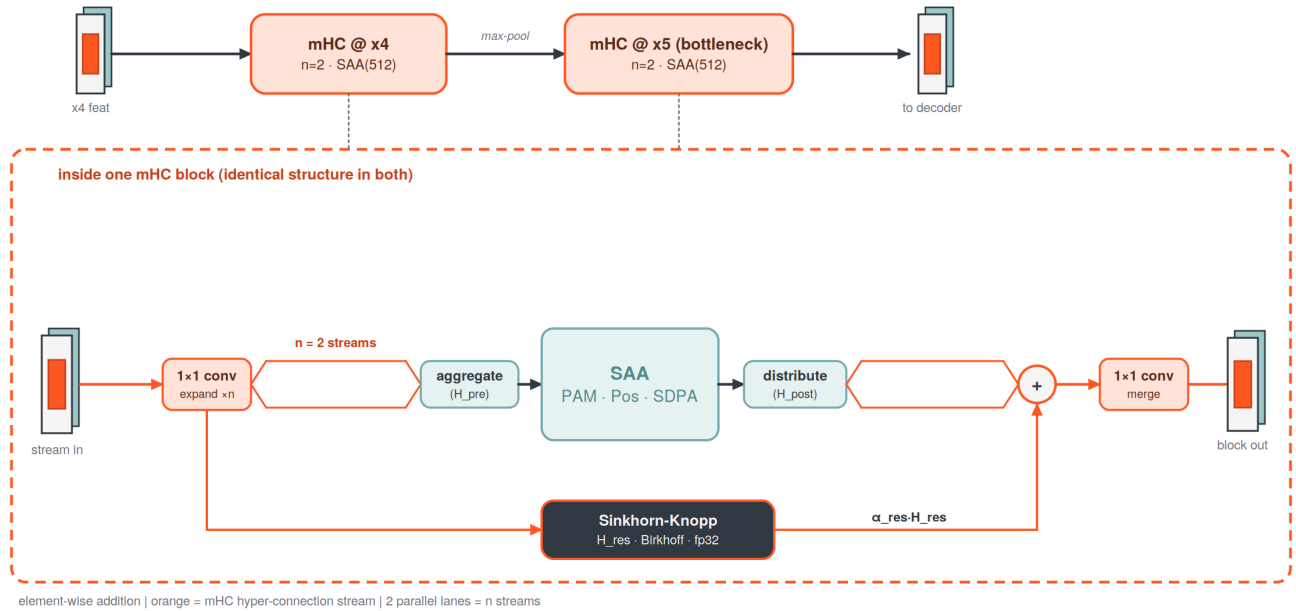


Fig. 3. Two-block HTAN_2 configuration. An mHC block is applied first at x_4 (512 channels, 32×32), and its output is max-pooled and passed to a second mHC block at x_5 (1024 channels, 16×16). Both blocks share the same internal structure shown in Fig. 2.

F. Objective Function

HTAN is trained using a combination of binary cross-entropy loss and Dice loss. Binary cross-entropy encourages accurate pixel-wise classification, while Dice loss directly optimizes the overlap between the predicted segmentation mask and the ground-truth mask. This combination is suitable for binary medical image segmentation tasks, where the foreground region may occupy only a small portion of the image.

The total loss is defined as:

$$\mathcal{L} = 0.5 \mathcal{L}_{\text{BCE}} + 0.5 \mathcal{L}_{\text{Dice}}. \quad (18)$$

The Dice loss is given by:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon}, \quad (19)$$

where p_i is the predicted probability for pixel i , y_i is the corresponding ground-truth label, and ϵ is a smoothing constant used to avoid division by zero.

IV. EXPERIMENTS

A. Datasets

We evaluate HTAN on three medical image segmentation benchmarks: ISIC-2018 skin lesion segmentation, GlaS gland segmentation, and Data Science Bowl 2018 nuclei segmentation. These datasets cover different segmentation targets, including skin lesions, glands, and cell nuclei, allowing us to evaluate whether the proposed mHC-enhanced bottleneck improves TransAttUNet across different medical segmentation settings.

1) *ISIC-2018 Skin Lesion Segmentation*: The ISIC-2018 dataset is used to evaluate skin lesion segmentation performance. This dataset contains dermoscopic images with corresponding lesion masks. Skin lesion segmentation is challenging because lesion boundaries can be irregular, low-contrast, and visually similar to surrounding healthy skin. In our experiments, ISIC-2018 is used as the main large-scale benchmark for comparing HTAN with reproduced U-Net, DoubleU-Net, TransAttUNet_R, and borrowed transformer-based baselines.

2) *GlaS Gland Segmentation*: The GlaS dataset is used to evaluate gland segmentation performance. It contains histology images with gland annotations. Compared with ISIC-2018, GlaS has a smaller training set, which makes the dataset sensitive to training stability and model generalization. For this reason, all GlaS experiments are trained for 150 epochs.

3) *Data Science Bowl 2018 Nuclei Segmentation*: The Data Science Bowl 2018 dataset is used to evaluate nuclei segmentation performance. This dataset contains microscopy images with nuclei masks. Nuclei segmentation requires accurate separation of small and often clustered objects. The task is relatively saturated for strong U-shaped baselines, making it useful for analyzing whether HTAN improves over its direct TransAttUNet_R baseline even when overall performance is already high.

B. Experimental Settings

1) *Baseline Models*: We compare HTAN against reproduced baselines and results borrowed from prior papers. Reproduced models are marked with *, while borrowed or original paper results are marked with †. The reproduced baselines include U-Net, DoubleU-Net, and TransAttUNet_R. The borrowed baselines include models such as Swin-UNet, SegFormer, MCTrans, Attention U-Net, ResUNet, MedT, KiU-Net, FANet, and Channel-UNet, depending on dataset availability.

TransAttUNet_R is the direct baseline for HTAN because HTAN is built by integrating Manifold-Constrained Hyper-Connections into the TransAttUNet bottleneck while preserving the TransAttUNet_R decoder design. DoubleU-Net is reported separately because the reproduced DoubleU-Net baseline uses a frozen pretrained ImageNet VGG16 encoder.

2) *Implementation Details*: HTAN integrates mHC into the Self-Aware Attention bottleneck of TransAttUNet. The mHC module expands the bottleneck representation into n residual streams and constrains residual stream mixing through Sinkhorn-Knopp projection. We evaluate variants with one or two mHC blocks and with $n = 2$ or $n = 4$ residual streams. We also include an Hres-only ablation to isolate the effect of constraining the residual stream mixing matrix.

A key implementation detail is numerical stability in Sinkhorn-Knopp normalization. In the HTAN_2 $n = 4$ setting, mixed-precision training originally produced NaN failures due to fp16 overflow during Sinkhorn-Knopp projection. This was fixed by forcing the Sinkhorn-Knopp computation to FP32, regardless of the global mixed-precision context. Gradient clipping with a maximum norm of 5.0 is also used for HTAN models where needed to stabilize training.

3) *Training Configuration*: Table II summarizes the training configuration used for each dataset. For ISIC-2018, the reproduced baselines follow the original SGD-based protocol, while HTAN is trained using AdamW because the mHC Sinkhorn-Knopp projection benefits from adaptive optimization. For GlaS and Bowl, AdamW is used for all reproduced baselines and HTAN variants, providing a direct same-optimizer comparison.

TABLE II
TRAINING CONFIGURATIONS USED FOR ISIC-2018, GLAS, AND BOWL.

Parameter	ISIC Base.	ISIC HTAN	GlaS	Bowl
Optimizer	SGD	AdamW	AdamW	AdamW
LR	10^{-4}	10^{-4}	10^{-4}	10^{-4}
Weight decay	10^{-4}	0.01	0.01	0.01
Scheduler	StepLR	Cosine	Cosine	Cosine
Epochs	100	100	150	100
Batch size	4	4	4	4
Grad. clip	None	5.0	None/5.0	None/5.0
Seed	123	123	123	123
Image size	256^2	256^2	128^2	256^2
TTA	Yes	Yes	Yes	Yes

4) *Evaluation Metrics*: We evaluate segmentation performance using Dice coefficient, intersection over union (IoU),

accuracy (ACC), recall (REC), and precision (PRE). Dice is used as the primary metric because it measures the overlap between the predicted segmentation mask and the ground-truth mask. Higher values indicate better performance for all metrics.

C. Quantitative Results

Table III summarizes the main Dice improvements of HTAN over the reproduced TransAttUNet_R baseline. HTAN improves over TransAttUNet_R on all three datasets, with Dice gains of 1.05%, 3.57%, and 1.07% on ISIC-2018, GlaS, and Bowl, respectively.

TABLE III
HEADLINE DICE IMPROVEMENTS OVER REPRODUCED
TRANSATTUNET_R.

Dataset	TransAttUNet_R*	HTAN Best	Gain	Variant
ISIC-2018	89.27	90.32	+1.05	HTAN_2 $n = 2$
GlaS	88.10	91.67	+3.57	HTAN_1 $n = 2$
Bowl	91.07	92.14	+1.07	HTAN_1 $n = 2$

1) *Results on ISIC-2018:* Table IV reports the results on ISIC-2018. The reproduced TransAttUNet_R baseline achieves 89.27% Dice. All HTAN variants improve over this baseline. HTAN_1 Hres-only achieves 89.95% Dice, HTAN_1 $n = 2$ achieves 90.15%, HTAN_2 $n = 2$ achieves 90.32%, and HTAN_1 $n = 4$ achieves the highest Dice score of 90.46%. Although HTAN_1 $n = 4$ obtains the highest Dice, the preferred $n = 2$ setting provides a stronger efficiency-performance tradeoff.

TABLE IV
RESULTS ON ISIC-2018 SKIN LESION SEGMENTATION.

Model	Params	Dice	IoU	ACC	REC	PRE
U-Net*	17.3M	87.76	79.08	95.10	86.20	90.81
DoubleU-Net* (pretrained)	9.2M	86.42	77.25	94.35	87.27	87.68
Swin-UNet†	—	89.72	82.90	—	90.32	92.04
SegFormer†	—	90.24	83.60	—	91.12	92.10
MCTrans†	—	90.35	—	—	—	—
TransAttUNet_R*	41.3M	89.27	81.24	95.72	87.62	92.24
HTAN_1 Hres-only	67M	89.95	82.41	95.92	89.42	91.78
HTAN_1 $n = 2$	47M	90.15	82.71	96.07	89.12	92.36
HTAN_1 $n = 4$	75M	90.46	83.07	96.13	89.55	92.33
HTAN_2 $n = 4$	129M	90.06	82.62	96.07	88.75	92.77
HTAN_2 $n = 2$	61M	90.32	82.93	96.02	89.77	92.00

* Reproduced results. † Borrowed results.

2) *Results on GlaS:* Table V reports the results on GlaS. The reproduced TransAttUNet_R baseline achieves 88.10% Dice under the AdamW setting used for direct comparison. The original TransAttUNet_R result is also reported as 89.11% Dice for transparency. HTAN_1 $n = 2$ obtains the best Dice score of 91.67%, improving over the reproduced TransAttUNet_R baseline by 3.57%.

3) *Results on Data Science Bowl 2018:* Table VI reports the results on Data Science Bowl 2018. The reproduced TransAttUNet_R baseline achieves 91.07% Dice. HTAN_1 $n = 2$ achieves 92.14% Dice, improving over TransAttUNet_R* by 1.07%. DoubleU-Net* obtains the highest Dice score of 92.91%, but it uses a frozen pretrained ImageNet VGG16 encoder. Therefore, the main claim on Bowl is that HTAN improves its direct TransAttUNet_R* baseline.

TABLE V
RESULTS ON GLAS GLAND SEGMENTATION.

Model	Params	Dice	IoU	ACC	REC	PRE
ResUNet†	—	80.88	69.11	81.49	85.11	80.01
MedT†	—	81.82	69.61	—	—	—
Attention U-Net†	—	81.59	70.06	—	—	—
KiU-Net†	—	83.25	72.78	—	—	—
FANet†	—	84.67	74.30	—	—	—
Swin-UNet†	—	86.70	77.32	—	89.00	86.12
SegFormer†	—	87.36	79.71	—	85.56	86.53
TransAttUNet_R†	41.3M	89.11	81.13	89.02	90.08	88.95
U-Net*	17.3M	90.27	82.77	90.55	89.36	91.54
DoubleU-Net* (pretrained)	9.2M	90.85	83.53	90.90	90.89	90.99
TransAttUNet_R*	41.3M	88.10	79.10	87.35	94.28	82.86
HTAN_1 Hres-only	54M	91.05	83.96	91.04	91.59	90.63
HTAN_1 $n = 4$	61M	91.34	84.40	91.47	90.99	91.87
HTAN_2 $n = 2$	51M	91.50	84.65	91.58	92.21	90.99
HTAN_1 $n = 2$	47M	91.67	84.94	91.71	91.71	91.76

* Reproduced results. † Borrowed or original paper results.

TABLE VI
RESULTS ON DATA SCIENCE BOWL 2018 NUCLEI SEGMENTATION.

Model	Params	Dice	IoU	ACC	REC	PRE
FANet†	—	81.03	71.08	95.59	80.62	82.31
Channel-UNet†	—	87.55	79.75	96.27	90.70	87.86
ResUNet†	—	89.91	82.44	97.05	90.00	90.84
Attention U-Net†	—	90.83	—	—	—	91.61
DoubleU-Net†	—	91.33	84.07	—	64.07	94.96
TransAttUNet_R†	41.3M	91.62	84.98	97.46	91.85	91.93
TransAttUNet_R*	41.3M	91.07	83.81	96.94	91.90	90.56
HTAN_1 Hres-only	67M	91.23	84.07	97.03	89.54	93.26
U-Net*	17.3M	92.01	85.37	97.30	91.02	93.27
HTAN_1 $n = 4$	75M	92.11	85.49	97.28	91.84	92.61
HTAN_2 $n = 2$	61M	92.13	85.54	97.28	92.09	92.36
HTAN_1 $n = 2$	47M	92.14	85.55	97.31	91.74	92.74
DoubleU-Net* (pretrained)	9.2M	92.91	86.86	97.54	92.45	93.56

* Reproduced results. † Borrowed or original paper results.

D. Ablation Study

1) *Effect of Adding mHC:* The ablation study on ISIC-2018 shows that adding mHC-based bottleneck processing improves over the reproduced TransAttUNet_R baseline. TransAttUNet_R* achieves 89.27% Dice, while HTAN variants reach between 89.95% and 90.46% Dice. This supports the central claim that mHC improves TransAttUNet for medical image segmentation.

2) *Effect of Hres-only:* HTAN_1 Hres-only achieves 89.95% Dice on ISIC-2018, improving over TransAttUNet_R* by 0.68%. This indicates that constraining the residual stream mixing matrix alone contributes positively, but the full mHC formulation provides further improvement.

3) *Effect of Expansion Rate n :* The expansion rate controls the number of residual streams. On ISIC-2018, HTAN_1 $n = 2$ achieves 90.15% Dice with 47M parameters, while HTAN_1 $n = 4$ achieves 90.46% Dice with 75M parameters. Although $n = 4$ gives a slightly higher Dice score in the one-block setting, it introduces more parameters and shows diminishing returns. In the deeper two-block setting, HTAN_2 $n = 2$ achieves 90.32% Dice, while HTAN_2 $n = 4$ drops to 90.06% Dice despite using 129M parameters.

4) *Effect of One vs. Two mHC Blocks:* The comparison between HTAN_1 $n = 2$ and HTAN_2 $n = 2$ shows that adding a second mHC block improves ISIC Dice from 90.15% to 90.32%. However, the comparison between HTAN_2 $n = 2$

and HTAN_2 $n = 4$ shows that deeper mHC stacking combined with larger stream expansion can reduce efficiency and stability. Therefore, $n = 2$ provides the best practical operating point across the tested settings.

TABLE VII
ISIC-2018 ABLATION OF MHC DESIGN CHOICES.

Model	Params	Dice
TransAttUNet_R*	41.3M	89.27
HTAN_1 Hres-only	67M	89.95
HTAN_1 $n = 2$	47M	90.15
HTAN_2 $n = 2$	61M	90.32
HTAN_1 $n = 4$	75M	90.46

E. Qualitative Visual Results

Fig. 4 presents qualitative segmentation examples from ISIC-2018, GlaS, and Data Science Bowl 2018. Each row shows the input image, ground-truth mask, reproduced TransAttUNet_R* prediction, and HTAN_2 $n = 2$ prediction. Across the selected examples, HTAN produces masks that more closely follow the ground truth than the reproduced TransAttUNet_R* baseline.

The sample-wise Dice scores further support this visual comparison. On ISIC-2018, HTAN improves Dice from 0.884 to 0.964 and from 0.802 to 0.884 on the shown examples. On GlaS, HTAN improves Dice from 0.654 to 0.781 and from 0.583 to 0.839. On Data Science Bowl 2018, HTAN improves Dice from 0.767 to 0.840 and from 0.794 to 0.873. These qualitative examples are consistent with the quantitative results, where HTAN improves over the reproduced TransAttUNet_R* baseline across all three datasets.

F. Stability and Implementation Analysis

1) *Sinkhorn-Knopp Numerical Stability*: The mHC module relies on Sinkhorn-Knopp projection to constrain the residual mixing matrix. This projection is important for maintaining stable residual stream mixing, but it also introduces numerical sensitivity during training, especially when the number of streams and mHC blocks increases.

2) *FP32 Sinkhorn Projection*: In the HTAN_2 $n = 4$ setting, Sinkhorn-Knopp projection originally caused NaN failures under AMP/fp16 mixed-precision training. The issue was fixed by forcing the Sinkhorn-Knopp computation to FP32. This implementation detail is important because it allows constrained residual stream mixing to remain numerically stable during training.

3) *Gradient Clipping*: Gradient clipping is also used to improve HTAN stability. For HTAN models, a maximum gradient norm of 5.0 is used where needed. This helps stabilize training with mHC, especially when Sinkhorn-Knopp projection and multi-stream residual mixing are applied.

4) *$n = 2$ vs. $n = 4$ Stability*: The comparison between $n = 2$ and $n = 4$ shows that increasing the number of streams is not always beneficial. On ISIC-2018, HTAN_2 $n = 2$ achieves 90.32% Dice with 61M parameters, while HTAN_2 $n = 4$ achieves 90.06% Dice with 129M parameters.

This indicates that $n = 4$ can increase complexity without improving performance. Overall, $n = 2$ provides a more stable and efficient operating point.

G. Parameter Analysis

Table VIII summarizes the parameter counts of the reproduced baseline and HTAN variants. TransAttUNet_R* has 41.3M parameters. HTAN_1 $n = 2$ increases the parameter count moderately to 47M and improves Dice on all three datasets. In contrast, HTAN_2 $n = 4$ reaches 129M parameters on ISIC-2018 and performs worse than HTAN_2 $n = 2$, showing that larger stream expansion is not automatically beneficial.

TABLE VIII
PARAMETER COMPARISON OF TRANSATTUNET_R* AND HTAN VARIANTS.

Model	ISIC	GlaS	Bowl
TransAttUNet_R*	41.3M	41.3M	41.3M
HTAN_1 Hres-only	67M	54M	67M
HTAN_1 $n = 2$	47M	47M	47M
HTAN_1 $n = 4$	75M	61M	75M
HTAN_2 $n = 2$	61M	51M	61M
HTAN_2 $n = 4$	129M	–	–

H. Discussion

The experimental results support the central claim of this work: mHC improves TransAttUNet for medical image segmentation. Across ISIC-2018, GlaS, and Bowl, HTAN consistently improves over the reproduced TransAttUNet_R baseline. The largest improvement is observed on GlaS, where HTAN_1 $n = 2$ improves Dice from 88.10% to 91.67%. On ISIC-2018, HTAN improves over TransAttUNet_R* and reaches competitive performance against transformer-based baselines. On Bowl, the task is more saturated: U-Net* reaches 92.01% Dice and HTAN_1 $n = 2$ reaches 92.14%. Therefore, the relevant claim on Bowl is not that HTAN outperforms every model, but that it improves its direct TransAttUNet_R* baseline.

The ablation results show that the number of residual streams must be selected carefully. Although HTAN_1 $n = 4$ achieves the highest ISIC Dice score among HTAN variants, the larger $n = 4$ setting gives diminishing returns and becomes less attractive when combined with deeper mHC stacking. HTAN_2 $n = 4$ has substantially more parameters than HTAN_2 $n = 2$ but performs worse on ISIC-2018. These results support the conclusion that $n = 2$ is a strong practical setting for mHC-enhanced TransAttUNet.

Finally, the optimizer setting should be interpreted carefully. On ISIC-2018, the reproduced baselines follow the original SGD-based protocol, while HTAN uses AdamW because mHC training benefits from adaptive optimization. Therefore, the ISIC result should be viewed together with the GlaS and Bowl experiments, where AdamW is used for all reproduced models and HTAN variants. These same-optimizer comparisons provide the clearest evidence that mHC improves the TransAttUNet bottleneck.

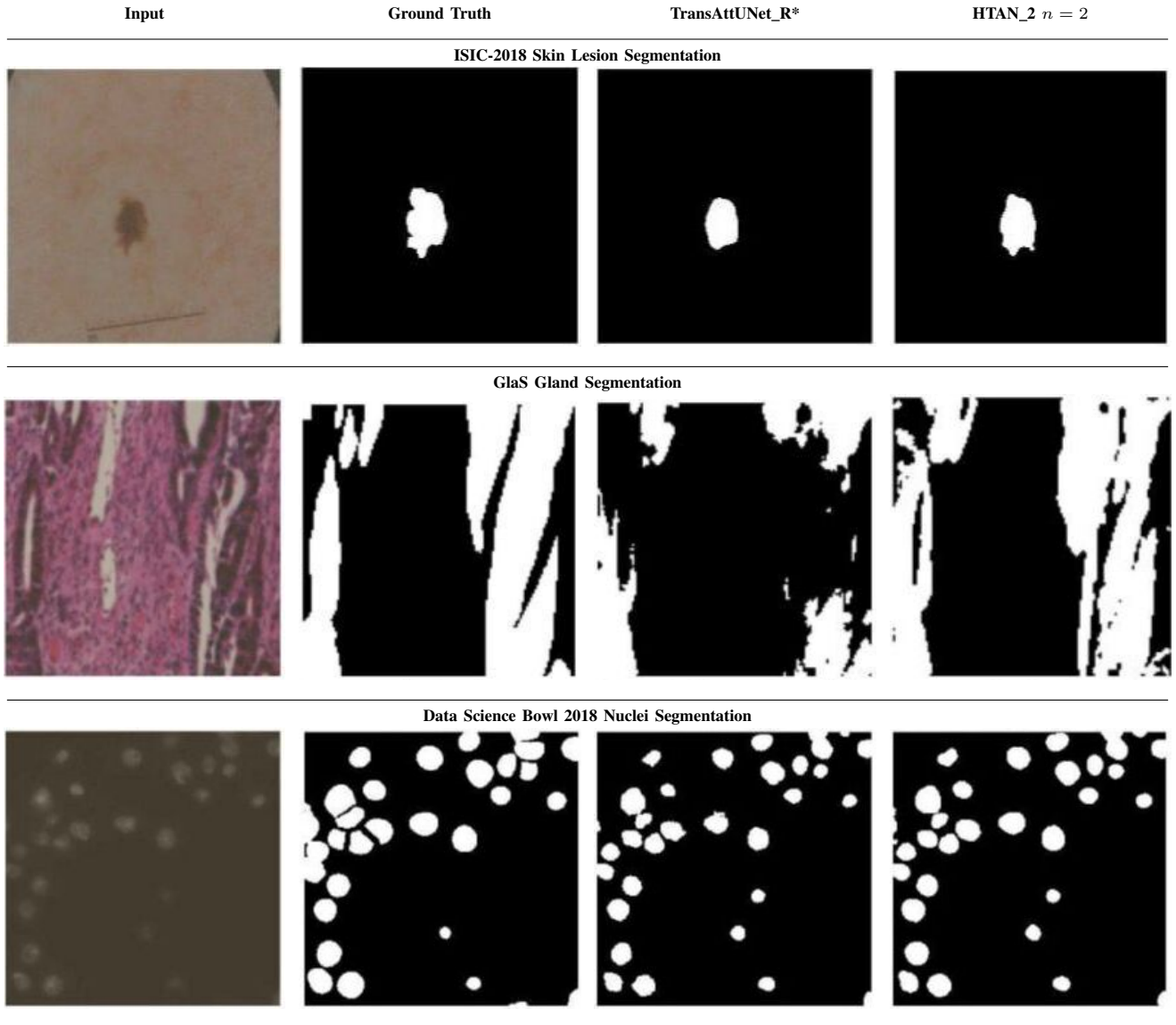


Fig. 4. Qualitative comparison across ISIC-2018, GlaS, and Data Science Bowl 2018. Columns show the input image, ground-truth mask, reproduced TransAttUNet_R* prediction, and HTAN_2 $n = 2$ prediction.

V. CONCLUSION

We presented HTAN, a medical image segmentation architecture that extends TransAttUNet with Manifold-Constrained Hyper-Connections at the Self-Aware Attention bottleneck. By expanding the bottleneck into n parallel residual streams with Sinkhorn-Knopp constrained mixing matrices, HTAN achieves richer feature representation while preserving training stability. We evaluated HTAN on three medical segmentation benchmarks: ISIC-2018 skin lesion segmentation, GlaS gland segmentation, and Data Science Bowl nuclei segmentation. Across all three datasets, HTAN consistently improves over the reproduced TransAttUNet_R baseline, achieving Dice gains of 1.05%, 3.57%, and 1.07%, respectively. Ablation studies confirm that $n = 2$ residual streams provide the best efficiency-performance tradeoff, and that the full mHC formulation

outperforms the Hres-only variant. These results demonstrate that manifold-constrained multi-stream bottleneck processing is an effective and stable way to improve attention-guided U-Net architectures for medical image segmentation. Future work will explore mHC integration at additional encoder levels with lightweight implementations and evaluate the approach on further segmentation tasks involving more complex anatomical structures.

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